

Computer Vision Practical Report:

DIRECT LINEAR TRANSFORMATION

1. AIM

To estimate the transformation matrix using the Direct Linear Transformation (DLT) algorithm from point correspondences, apply it to transform an image using OpenCV, display the original and transformed images, and save the result to disk.

2. PROCEDURE

- Load the image and collect corresponding source and destination point coordinates.
- Formulate the matrix equation $A \mathbf{h} = 0$ using the point pairs and compute the Homography matrix H using Singular Value Decomposition (SVD).
- Apply the computed matrix to transform the image using `cv2.warpPerspective()`.
- Display the original and transformed images side-by-side using Matplotlib.
- Save the transformed image using `cv2.imwrite()`.

3. PROGRAM CODE

```
import cv2
import numpy as np
image_path = r"C:\Users\adminuser\Desktop\Python CV\E15-IMG.jpg"

image = cv2.imread(image_path)

if image is None:
    print("ERROR: Image could not be loaded.")
    print("Check the image path:", image_path)
    exit()

print("Image loaded successfully.")
print("Image size:", image.shape)
src_points = np.float64([
```

```

    [100, 100],
    [500, 100],
    [500, 400],
    [100, 400]
])
dst_points = np.float64([
    [50, 150],
    [550, 80],
    [500, 450],
    [100, 500]
])

A = []

for (x, y), (u, v) in zip(src_points, dst_points):

    A.append([
        -x, -y, -1,
        0, 0, 0,
        u*x, u*y, u
    ])

    A.append([
        0, 0, 0,
        -x, -y, -1,
        v*x, v*y, v
    ])

A = np.array(A)

print("\nDLT Matrix A:")
print(A)

U, S, Vt = np.linalg.svd(A)

```

```
# Last row of Vt gives the solution
h = Vt[-1]

# Convert 9 values into a 3x3 matrix
H = h.reshape(3, 3)

# Normalize H
H = H / H[2, 2]
print("\nHomography Matrix using DLT:")
print(H)

height = 600
width = 600

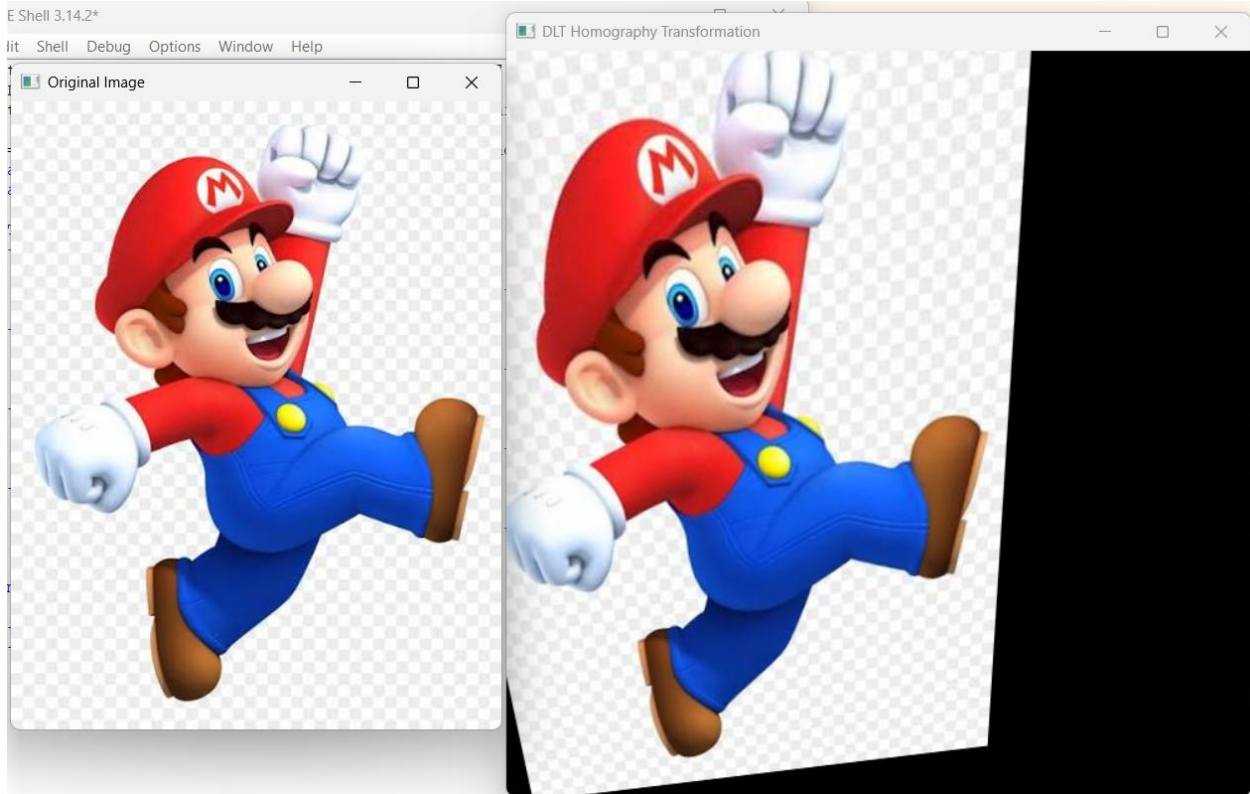
transformed_image = cv2.warpPerspective(
    image,
    H,
    (width, height)
)

cv2.imshow("Original Image", image)
cv2.imshow("DLT Homography Transformation", transformed_image)

cv2.waitKey(0)
cv2.destroyAllWindows()
cv2.imwrite("DLT_transformed.jpg", transformed_image)

print("\nTransformation completed successfully.")
print("Output saved as DLT_transformed.jpg")
```

4. OUTPUT



5. RESULT

The Python script executed successfully. The input image was loaded.